

Lecture 3

Causal Questions



Announcements

- ❖ Lab 0 today

- Grab your computer during the break if you didn't bring it today

Last Time

- ❖ Surveyed summary statistics (mean, median, min/max, etc.)
- ❖ Types of questions we can ask about a dataset
 - Descriptive
 - Comparative
 - Associative

Exercise

Imagine you have a dataset containing daily swipe data for the dining halls

- ❖ Variables:
 - date, dining hall, # swipes, featured item, semester week, coffee consumed

Formulate a descriptive, a comparative, and an associative question

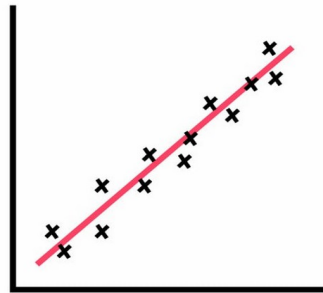
- ❖ Offer a hypothesis about what you expect to find

Today

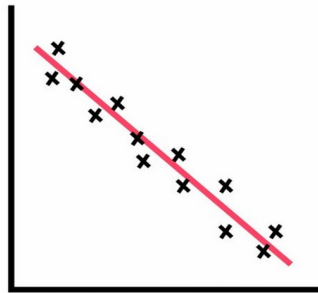
- ❖ Causal questions
- ❖ Model for thinking about causation

Associative Questions

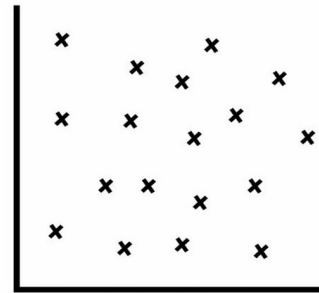
- ❖ Tell us how variables *correlate*
 - Positive correlation means as one increases the other increases
 - Negative correlation means as one increases the other decreases
- ❖ Measuring and examining correlations is a major part of data analysis



Positive
Correlation



Negative
Correlation



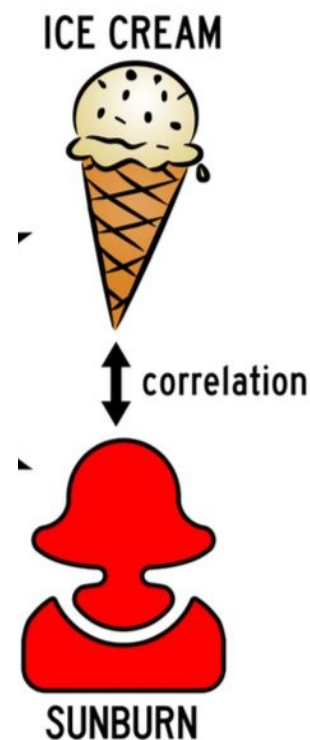
No
Correlation

Interpreting Association

- ❖ Suppose variables X and Y are associated
- ❖ Means: knowing X tells you something about Y and vice versa

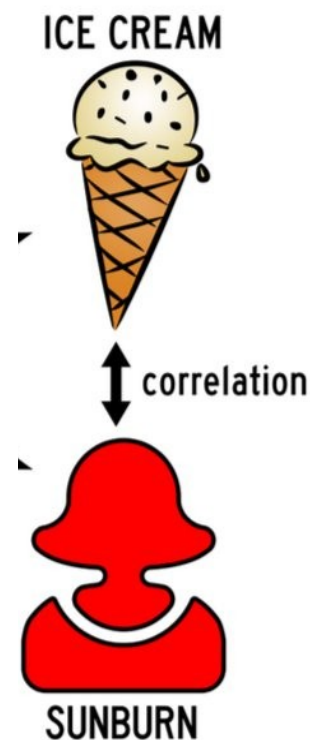
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- ❖ Example: sunburns & ice cream
 - Ice cream sales correlate with # of reported sunburns



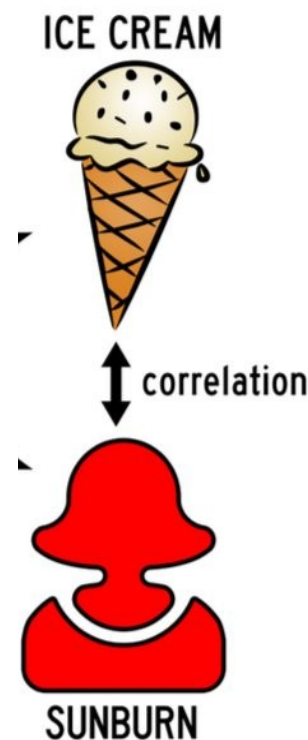
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- ❖ Example: sunburns & ice cream
 - Ice cream sales correlate with # of reported sunburns
- ❖ Can we conclude that consuming more ice cream causes more sunburns?
 - Not clear yet what exactly this means
 - Not from correlation alone



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 - Intervention: If we could change the value of X (but not change anything else), would Y also change?
- ❖ Causal questions can be very difficult to answer well!

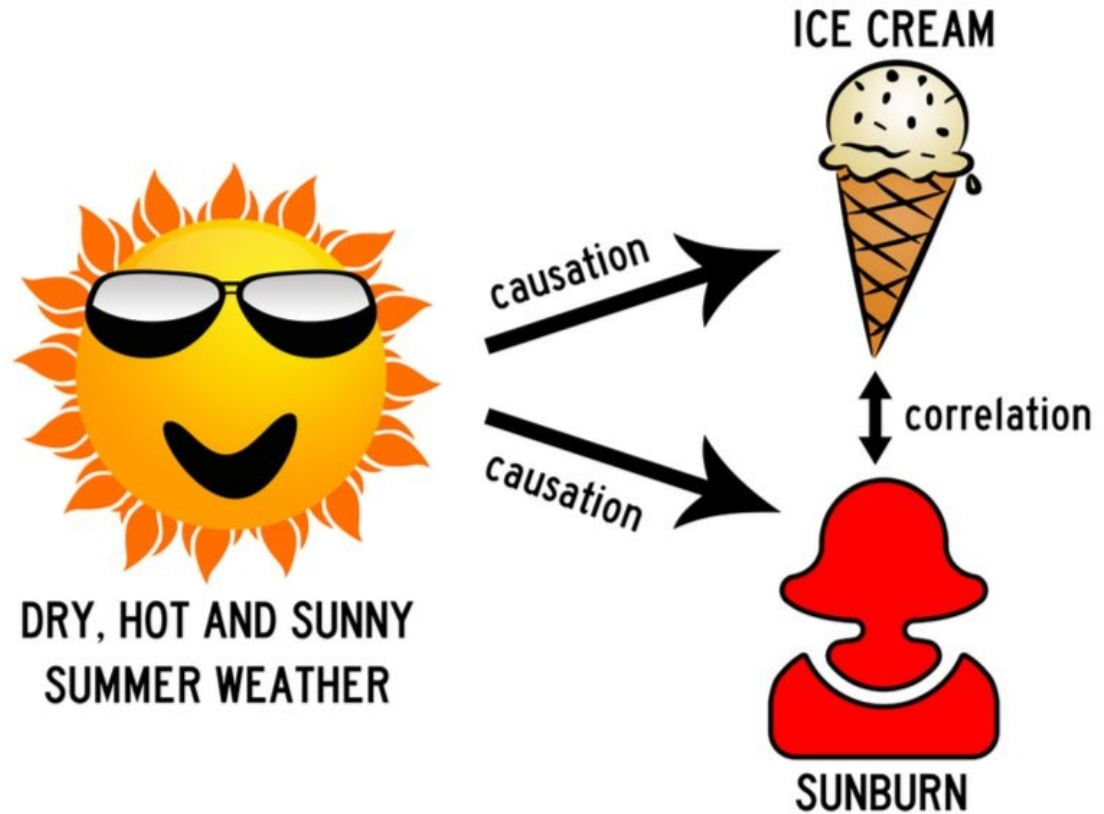


Confounding

- ❖ Some correlated variables have common causes
- ❖ We call those confounders

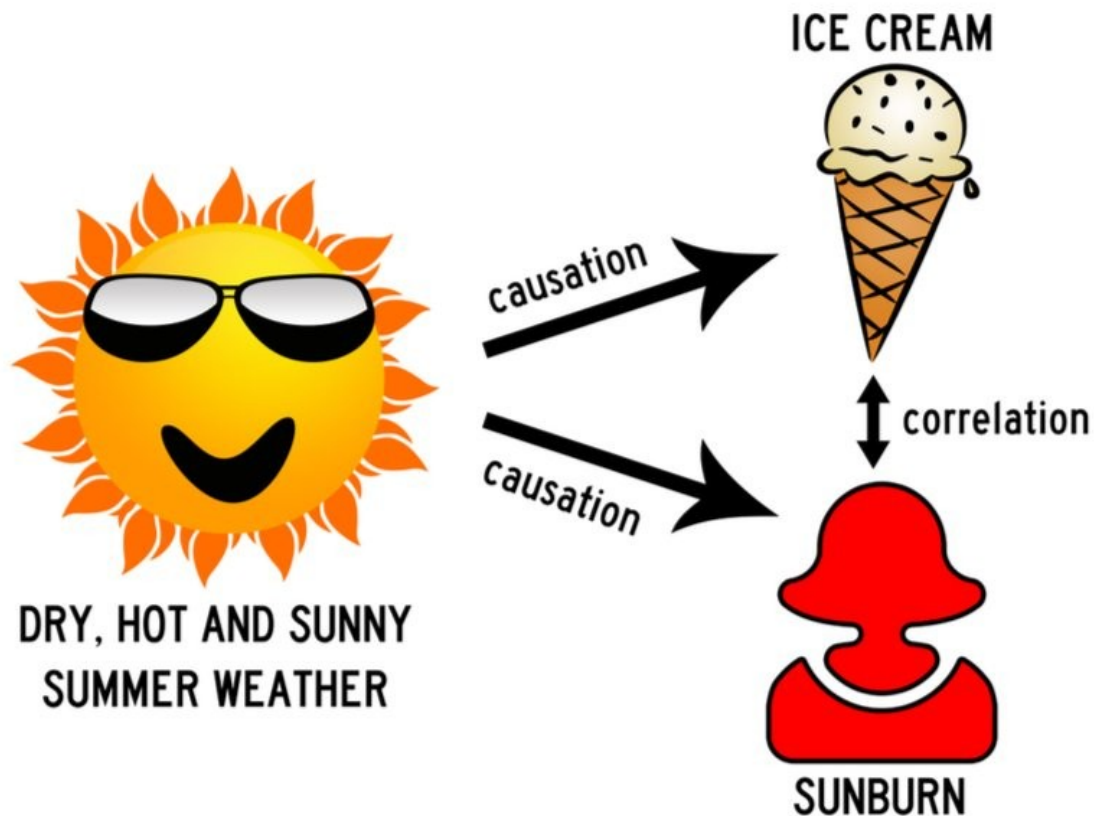
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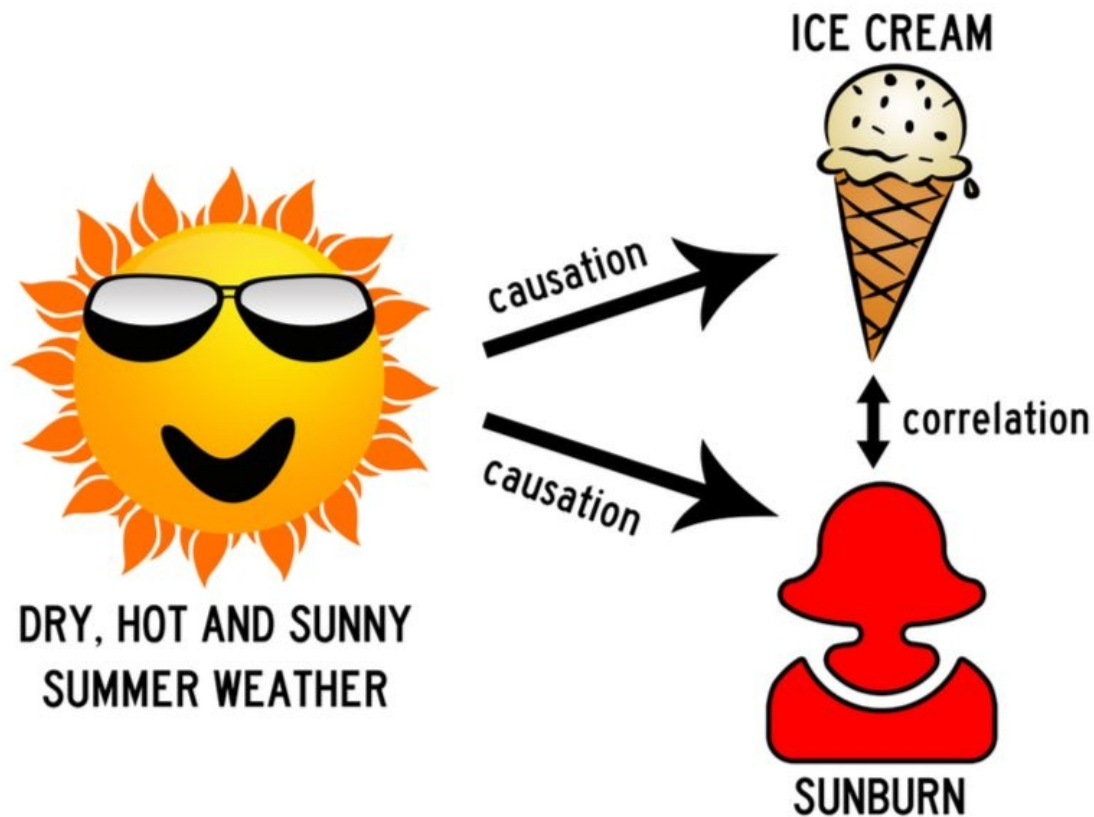
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- ❖ How pervasive is confounding?



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- ❖ Sunny weather causes both ice cream consumption and sunburns
- ❖ How pervasive is confounding?
 - Quite pervasive!



Closing Thoughts

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Closing Thoughts

- ❖ Causal claims are pervasive in the social and behavioral sciences
- ❖ But confounders may be everywhere!
- ❖ How might we go about answering causal questions?

Next Time

- ❖ Read CIT Ch. 2
- ❖ How to we answer causal questions?
- ❖ Famous example: Cholera Investigation in 1850s London